

students	watch	youtube	write	comments

$TF = \frac{\text{Number of times a word "X" appears in a Document}}{\text{Number of words present in a Document}}$

$IDF = \log \left(\frac{\text{Number of Documents present in a Corpus}}{\text{Number of Documents where word "X" has appeared}} \right)$

$TF \cdot IDF = TF \cdot IDF$

Document	Text
D1	students watch youtube
D2	youtube watch youtube
D3	students write comments
D4	youtube write comments

Term frequency : word = term

D1 : students watch youtube
N.W = 3

	TF	TF	IDF	TF * IDF
students	1/3	0.33	0.69	0.207
watch	1/3	0.33	0.69	0.207
youtube	1/3	0.33	0.28	0.084

D2 : youtube watch youtube
N.W = 3

TF	IDF
0.33	0.28

youtube	4/3	0.3
watch	1/3	0.3

D3 students write comments

row = 3

	T.F	
students	1/3	0.3
write	1/3	0.3
comments	1/3	0.3

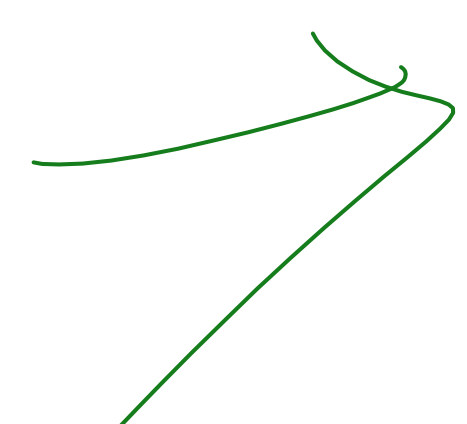
D4 youtube write comments

row = 3

	T.F	
youtube	1/3	0.3
write	1/3	0.3
comments	1/3	0.3

D4 youtube write comments I.D.F

⇒ 3



$$\text{youtube} = \frac{4}{3}$$

$$\text{write} = \frac{4}{3}$$

$$\text{comments} = \frac{4}{3}$$

0.3

0.3

0.3

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$$TF \cdot IDF = TF * IDF$$

students	watch	youtube	write	comments
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Document	Text
D1	students watch youtube
D2	youtube watch youtube
D3	students write comments
D4	youtube write comments

IDF

IDF:

$$\text{Students} \rightarrow \log\left(\frac{4}{2}\right) \rightarrow 0.69$$

$$, 4 \rightarrow 0.69$$

$$\text{watch} \rightarrow \log\left(\frac{1}{2}\right)$$

$$\text{youtube} \rightarrow \log\left(\frac{4}{3}\right) \rightarrow 0.28$$

$$\text{write} \rightarrow \log\left(\frac{4}{2}\right) \rightarrow 0.69$$

$$\text{comment} \rightarrow \log\left(\frac{4}{2}\right) \rightarrow 0.69$$

TF $\rightarrow$
document
ଏକ ଡକ୍ୟୁମେଣ୍ଟର
value

IDF

୧୨- word

ନାମର IDF

TF*DP